3.4.1 Force, energy and momentum

3.4.1.1 Scalars and vectors

Content	Opportunities for skills development
Nature of scalars and vectors. Examples should include: velocity/speed, mass, force/weight, acceleration, displacement/distance. Addition of vectors by calculation or scale drawing. Calculations will be limited to two vectors at right angles. Scale drawings may involve vectors at angles other than $90°$. Resolution of vectors into two components at right angles to each other. Examples should include components of forces along and perpendicular to an inclined plane. Problems may be solved either by the use of resolved forces or the use of a closed triangle. Conditions for equilibrium for two or three coplanar forces acting at a point. Appreciation of the meaning of equilibrium in the context of an object at rest or moving with constant velocity.	**MS 0.6, 4.2, 4.4, 4.5 / PS 1.1** Investigation of the conditions for equilibrium for three coplanar forces acting at a point using a force board.

3.4.1.2 Moments

Content	Opportunities for skills development
Moment of a force about a point. Moment defined as *force* × *perpendicular distance from the point to the line of action of the force*. Couple as a pair of equal and opposite coplanar forces. Moment of couple defined as *force* × *perpendicular distance between the lines of action of the forces*. Principle of moments. Centre of mass. Knowledge that the position of the centre of mass of uniform regular solid is at its centre.	

3.4.1.3 Motion along a straight line

Content	Opportunities for skills development
Displacement, speed, velocity, acceleration. $v = \frac{\Delta s}{\Delta t}$ $a = \frac{\Delta v}{\Delta t}$ Calculations may include average and instantaneous speeds and velocities. Representation by graphical methods of uniform and non-uniform acceleration. Significance of areas of velocity–time and acceleration–time graphs and gradients of displacement–time and velocity–time graphs for uniform and non-uniform acceleration eg graphs for motion of bouncing ball. Equations for uniform acceleration: $v = u + at$ $s = \left(\frac{u+v}{2}\right)t$ $s = ut + \frac{at^2}{2}$ $v^2 = u^2 + 2as$ Acceleration due to gravity, g.	**MS 3.6, 3.7 / PS 1.1, 3.1** Distinguish between instantaneous velocity and average velocity. **MS 3.5, 3.6** Measurements and calculations from displacement–time, velocity–time and acceleration–time graphs. **MS 0.5, 2.2, 2.3, 2.4** Calculations involving motion in a straight line.
Required practical 3: Determination of g by a freefall method.	**MS 0.3, 1.2, 3.7 / AT d** Students should be able to identify random and systematic errors in the experiment and suggest ways to remove them. **MS 3.9** Determine g from a graph.

3.4.1.4 Projectile motion

Content	Opportunities for skills development
Independent effect of motion in horizontal and vertical directions of a uniform gravitational field. Problems will be solvable using the equations of uniform acceleration. Qualitative treatment of friction. Distinctions between static and dynamic friction will not be tested. Qualitative treatment of lift and drag forces. Terminal speed. Knowledge that air resistance increases with speed. Qualitative understanding of the effect of air resistance on the trajectory of a projectile and on the factors that affect the maximum speed of a vehicle.	**PS 2.2, 3.1** Investigation of the factors that determine the motion of an object through a fluid.

3.4.1.5 Newton's laws of motion

Content	Opportunities for skills development
Knowledge and application of the three laws of motion in appropriate situations. $F = ma$ for situations where the mass is constant.	**PS 4.1 / MS 0.5, 3.2 / AT a, b, d** Students can verify Newton's second law of motion. **MS 4.1, 4.2** Students can use free-body diagrams.

3.4.1.6 Momentum

Content	Opportunities for skills development
$momentum = mass \times velocity$ Conservation of linear momentum. Principle applied quantitatively to problems in one dimension. Force as the rate of change of momentum, $F = \frac{\Delta(mv)}{\Delta t}$ Impulse = change in momentum $F\Delta t = \Delta(mv)$, where F is constant. Significance of the area under a force–time graph. Quantitative questions may be set on forces that vary with time. Impact forces are related to contact times (eg kicking a football, crumple zones, packaging). Elastic and inelastic collisions; explosions. Appreciation of momentum conservation issues in the context of ethical transport design.	**MS 2.2, 2.3** Students can apply conservation of momentum and rate of change of momentum to a range of examples.

3.4.1.7 Work, energy and power

Content	Opportunities for skills development
Energy transferred, $W = Fs\cos\theta$ $rate\ of\ doing\ work = rate\ of\ energy\ transfer,\ \ P = \frac{\Delta W}{\Delta t} = Fv$ Quantitative questions may be set on variable forces. Significance of the area under a force–displacement graph. $efficiency = \frac{useful\ output\ power}{input\ power}$ Efficiency can be expressed as a percentage.	**MS 0.3 / PS 3.3, 4.1 / AT a, b, f**. Investigate the efficiency of an electric motor being used to raise a mass through a measured height. Students should be able to identify random and systematic errors in the experiment and suggest ways to remove them.

3.4.1.8 Conservation of energy

Content	Opportunities for skills development
Principle of conservation of energy. $\Delta E_p = mg\Delta h$ and $E_k = \frac{1}{2}mv^2$ Quantitative and qualitative application of energy conservation to examples involving gravitational potential energy, kinetic energy, and work done against resistive forces.	**MS 0.4, 2.2** Estimate the energy that can be derived from food consumption.